MILITARY STANDARD

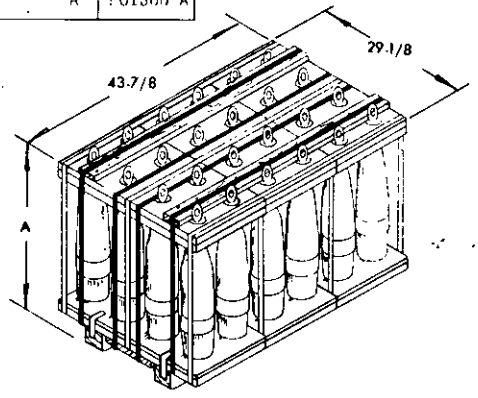
MIL-STD-1320-238
(NAVY)
10 MARCH 1989

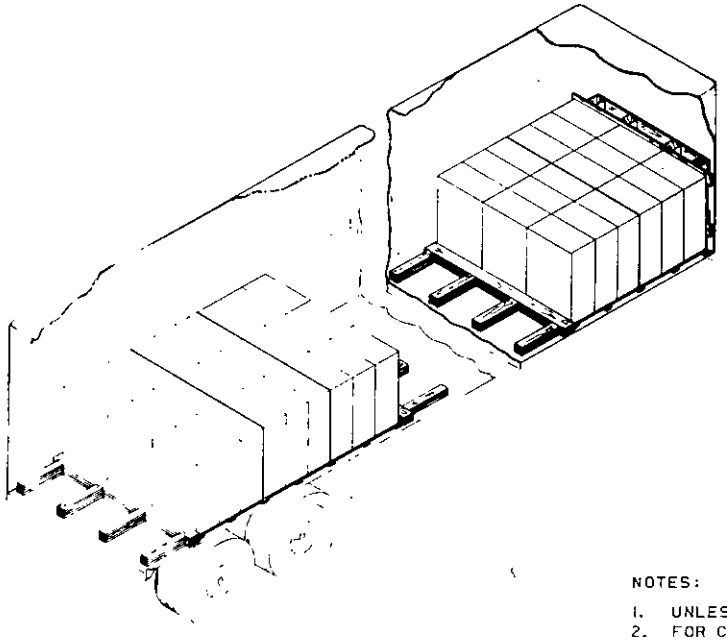
TRUCKLOADING

AMPHIBIOUS UNIT LOAD
155MM PROJECTILES, LONG ROUND

UNIT LOAD DATA

DODIC/NALC	D501/D503 D509/D514	D502	D515	D563	D579	D594
NUMBER OF PROJECTILES PER PALLET	8	8	8	8	8	8
NUMBER OF PALLETS PER UNIT LOAD	3	3	3	3	3	3
NUMBER OF PROJECTILES PER UNIT LOAD	24	24	24	24	24	24
GROSS WEIGHT OF 1 PALLET (LBS)	882	886	882	874	830	799
GROSS WEIGHT OF 1 UNIT LOAD (LBS)	2676	2688	2676	2652	2520	2427
PALLETIZATION DRAWING 19200 -	8837839	8837839	8837839	8837839	9327883	8837839
DIMENSION A (UNIT LOAD HEIGHT)	39-3/8	39-3/8	38-3/4	39-3/8	38-3/4	40-7/8
CUBE (CU FT)	29.1	29.1	28.7	29.1	28.7	30.2
FTL WEIGHT (LBS)	40,140	40,320	40,140	39,780	37,800	36,405
HAZARD CLASSIFICATION (DOT)	A	A	A	A	A	POISON A





INDEX

GENERAL NOTES.....2
FTL VAN.....3
LTL VAN.....4
UNITIZING.....5

AMSC NO. - N/A

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE: DISTRIBUTION IS UNLIMITED.

NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
			APPROVAL	

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY.

E. Ronaldi 3-9-89
SIGNATURE TDA, WPNSTA EARLE DATE

R R Caringi 3-10-89
SIGNATURE NAVSEASYSCOM BY DIRECTION DATE

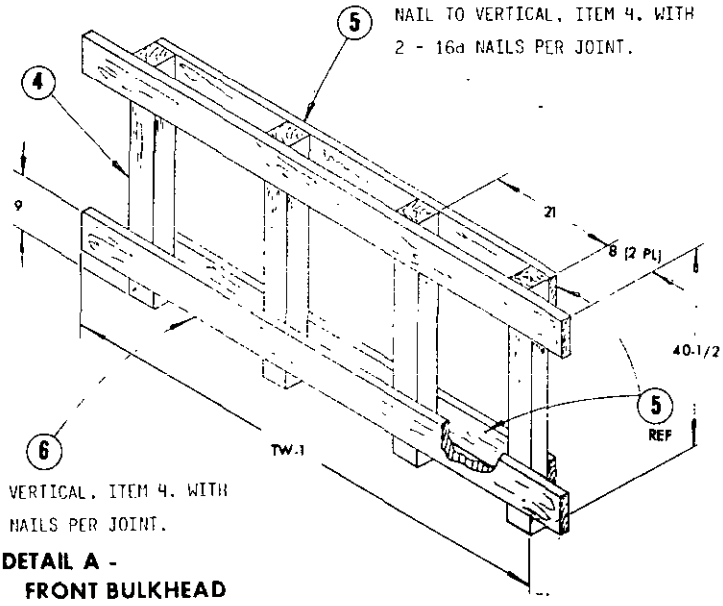
ORIGINATOR S.R. Petri 14 Feb 1989
DATE

WPNSTA EARLE, NEW JERSEY

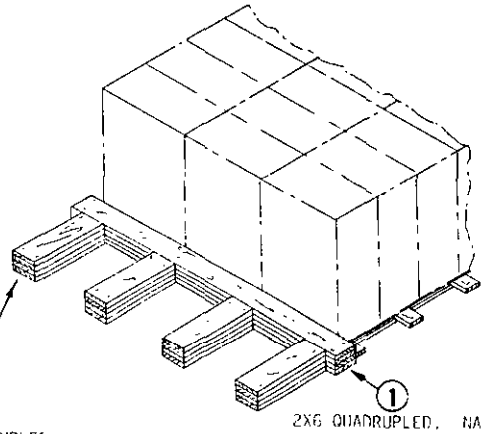
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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING 155MM AMMUNITION AMPHIBIOUS UNIT LOADS ASSEMBLED PER DRAWING MWHC 10271 OR AS SHOWN IN THE UNITIZING DETAIL. INDIVIDUAL PALLETS, WHEN NOT REQUIRED AS UNIT LOADS, CAN BE TRUCKLOADED PER DARCOM DRAWING 19-48-4019-11PE1000.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT VAN TRAILERS WITH NAILABLE FLOORS. ALL TRAILERS SHALL HAVE THEIR TANDEM AXLES LOCATED AT THE EXTREME REAR OF THE TRAILER (WESTERN POSITION).
3. THE SIDE BLOCKING, DETAILS D AND E, ARE INSTALLED AFTER THE UNIT LOADS ARE IN PLACE AND BEFORE THE CROSSMEMBERS ARE INSTALLED.
4. *THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE* RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS.
5. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH APPROPRIATE HAZARD CLASSIFICATION PLACARDS TO BOTH SIDES, FRONT, AND BACK OF THE TRAILER.
6. *UNLESS OTHERWISE SPECIFIED, MATERIALS AND PROCEDURES SHALL BE IN ACCORDANCE WITH THE GENERAL* TRUCKLOADING DOCUMENT, MIL-STD-1320.



DETAIL A - FRONT BULKHEAD

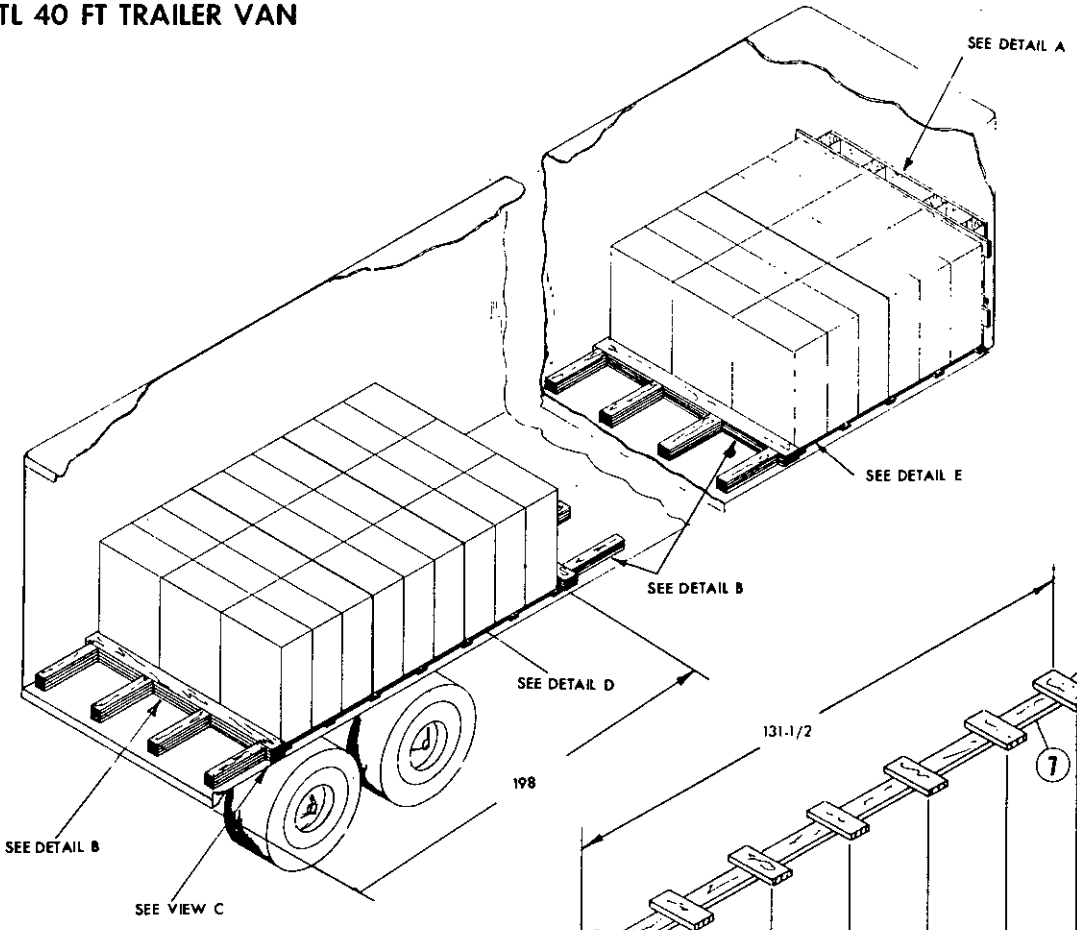


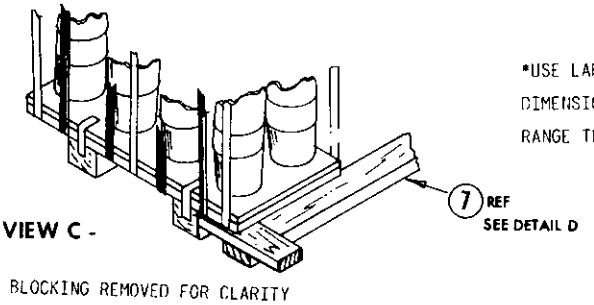
DETAIL B - (2) 2X6 QUADRUPLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 7 - 16d NAILS 2ND LAYER TO 1ST, 3RD TO 2ND, AND 4TH TO 3RD IN LIKE MANNER.

(1) 2X6 QUADRUPLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 1 - 16d NAIL EVERY 6 - 8 INCHES. NAIL 2ND LAYER TO 1ST, 3RD TO 2ND, AND 4TH TO 3RD IN LIKE MANNER.

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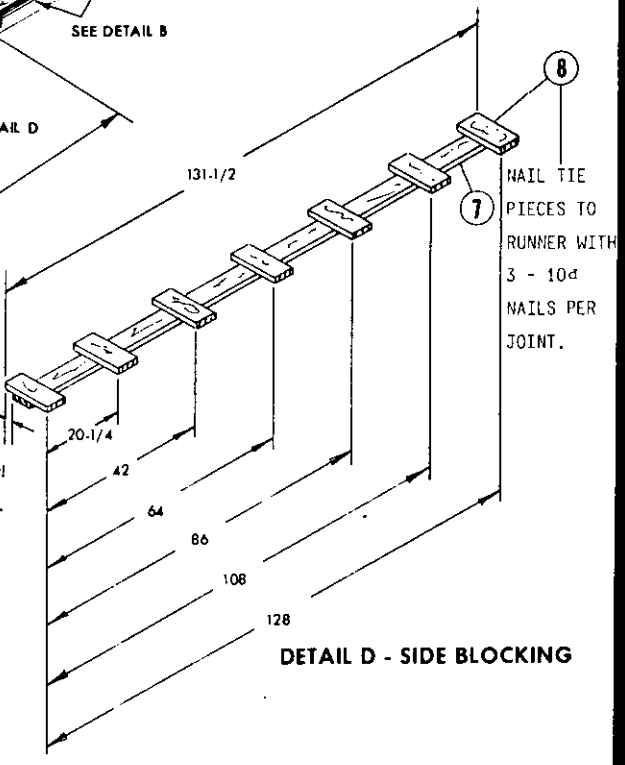
FTL 40 FT TRAILER VAN



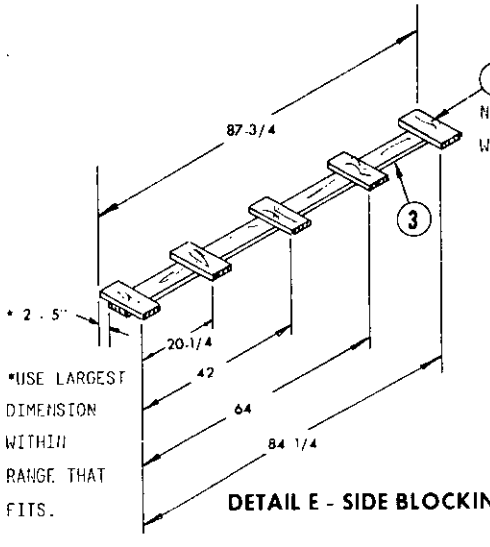


VIEW C -

BLOCKING REMOVED FOR CLARITY



DETAIL D - SIDE BLOCKING



DETAIL E - SIDE BLOCKING

*TW = TRAILER WIDTH (INSIDE)

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
24	8	SPACER	WOOD, NOTE 6	2 X 4 X TO SUIT
2	7	RUNNER, LONG	WOOD, NOTE 6	2 X 4 X 131-1/2
2	6	AFT CROSSMEMBER	WOOD, NOTE 6	2 X 6 X *TW-1
2	5	FWD CROSSMEMBER	WOOD, NOTE 6	2 X 6 X *TW-17
4	4	VERTICAL	WOOD, NOTE 6	4 X 4 X 40-1/2
2	3	RUNNER, SHORT	WOOD, NOTE 6	2 X 6 X 87-3/4
48	2	BACKUP CLEATS	WOOD, NOTE 6	2 X 6 X 24
12	1	CROSSMEMBER	WOOD, NOTE 3,6	2 X 6 X *TW-1

LIST OF MATERIALS PAGE 3

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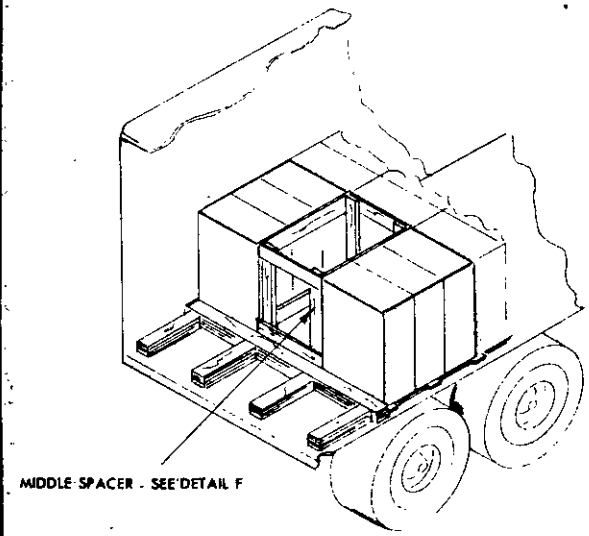
LTL 40 FT TRAILER VAN

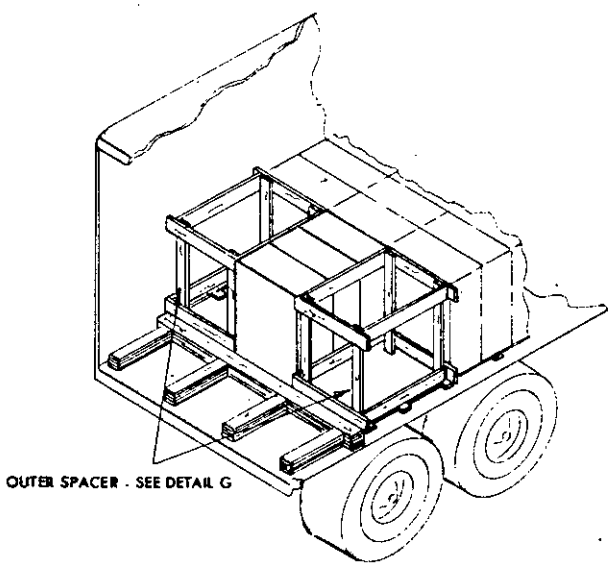
A. FOR LESS THAN TRUCKLOADS, REMOVE INDIVIDUAL LOADS FROM THE REAR OF THE VAN.

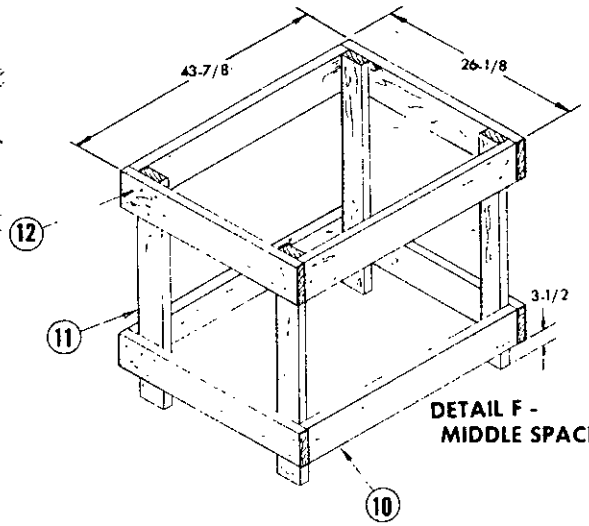
B. FOR SHIPMENTS OF 3, 6, 9, OR 12 UNIT LOADS, REMOVE ENTIRE ROWS.

C. FOR SHIPMENTS OF 2, 5, 8, 11, OR 14 UNIT LOADS, REPLACE INNER UNIT LOAD OF REARMOST ROW WITH ONE MIDDLE SPACER, DETAIL F.

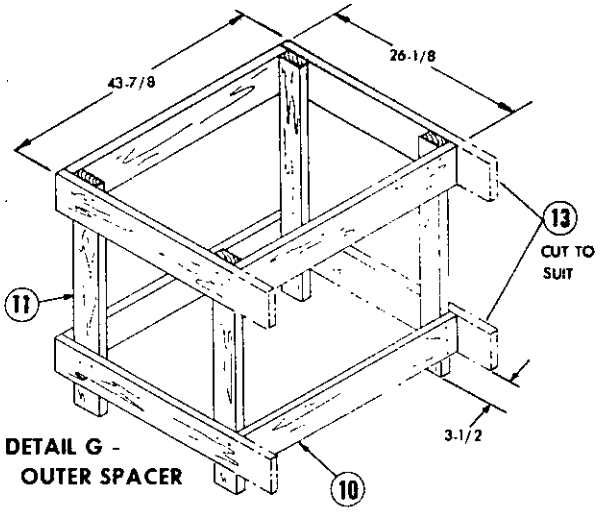
D. FOR SHIPMENTS OF 1, 4, 7, 10, OR 13 UNIT LOADS, REPLACE BOTH OUTER UNIT LOADS OF REARMOST ROW WITH OUTER SPACERS, DETAIL G.







DETAIL F - MIDDLE SPACER



DETAIL G - OUTER SPACER

SPACER ASSEMBLY INSTRUCTIONS

A. NAIL VERTICAL BOARD, ITEM 11, TO LATERAL BOARD, ITEM 12, OR SIDE BOARD, ITEM 13, WITH 3 - 10d NAILS PER JOINT.

B. SIDE NAIL LONGITUDINAL BOARDS, ITEM 10, TO VERTICAL BOARDS, ITEM 11, WITH 2 - 10d NAILS PER JOINT.

C. END NAIL LATERAL BOARD, ITEM 12, OR SIDE BOARD, ITEM 13, TO LONGITUDINAL BOARD, ITEM 10, WITH 1 - 10d NAIL PER JOINT.

D. FOR OUTER SPACERS, CUT SIDE BOARDS, ITEM 13, TO OBTAIN A SNUG FIT AGAINST TRAILER WALLS.

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
4	13	SIDE BOARD	DETAIL G	2 X 6 X TO SUIT
4	12	LATERAL BOARD	DETAIL F	2 X 6 X 29-1/8
4	11	VERTICAL BOARD	DETAIL F,G	2 X 4 X 40-1/2
4	10	LONGITUDINAL BOARD	DETAIL F,G	2 X 6 X 40-7/8

LIST OF MATERIALS

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UNITIZING PROCEDURE

A. PLACE PALLET LOADS, ITEM 1, TOGETHER WITH LONG SIDES TOGETHER AS SHOWN.

B. PLACE INNER AND OUTER STRAPS, ITEMS 5 AND 4, UNDER THE PALLETS.

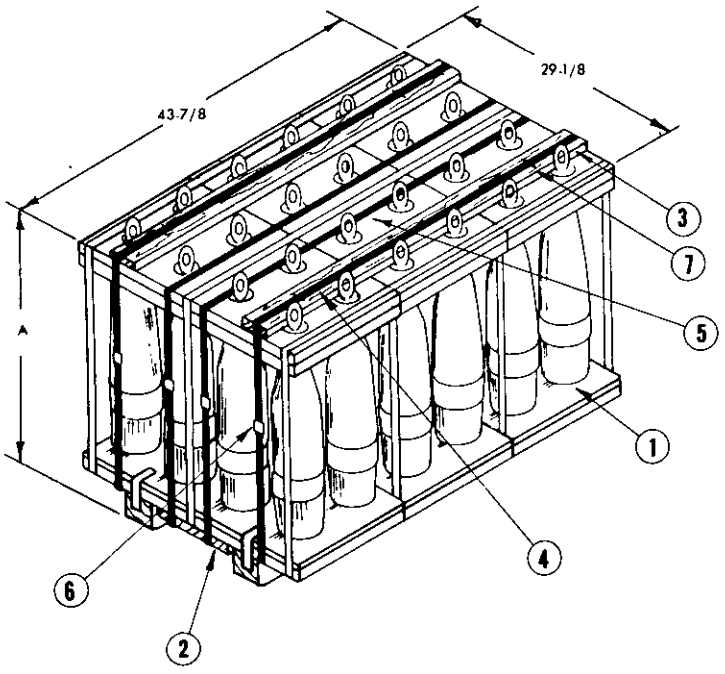
C. PLACE SPACER BOARD, ITEM 3, INSIDE AND AGAINST THE OUTER PROJECTILE LUGS.

D. POSITION OUTER STRAPPING, ITEM 4, AROUND LOAD AND OVER SPACER BOARD, ITEM 3. TENSION AND DOUBLE NOTCH SEAL WITH ITEM 6.

E. STAPLE OUTER STRAP, ITEM 4, TO SPACER BOARD, ITEM 3, WITH STAPLES, ITEM 7, IN THE LOCATION SHOWN.

F. PLACE SUPPORT BOARD, ITEM 2, OVER INNER STRAPS, ITEM 5.

G. POSITION INNER STRAPS, ITEM 5, AROUND LOAD. TENSION UNTIL SUPPORT BOARD, ITEM 2, IS FIRMLY AGAINST PALLETS. DOUBLE NOTCH SEAL WITH ITEM 6.



DEUNITIZING PROCEDURE

A. CUT STRAPS, ITEMS 4 AND 5.

B. REMOVE BOARDS, ITEMS 2 AND 3.

C. KEEP PALLETIZED PROJECTILES TOGETHER UNTIL NEEDED.

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
4	7	STAPLE	NOTE 6	3/4
4	6	SEAL	STEEL, NOTE 6	3/4 X .035
2	5	INNER STRAP	STEEL, NOTE 6	3/4 X .035 X 15 FT
2	4	OUTER STRAP	STEEL, NOTE 6	3/4 X .035 X 15 FT
2	3	SPACER BOARD	WOOD, NOTE 6	2 X 4 X 43-3/4
1	2	SUPPORT BOARD	WOOD, NOTE 6	2 X 10 X 43-3/4
3	1	PALLETIZED PROJECTILES	9327883/8837839	29-1/8 X 14-5/8 X A

LIST OF MATERIALS

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N803)

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